

PROJECT REPORT

Seasonal Agriculture Performance Analysis

*A Data Analytics study of yield, profitability, water-use efficiency and risk across 4,000 farms,
three growing seasons and eight Indian states*

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Prepared by: Vaibhav Deepak Mane

Shivajirao S. Jondhale College of Engineering

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Executive Summary

This project analyzes 4,000 farm records spanning three agricultural seasons — Kharif, Rabi and Zaid — across eight Indian states and eight crops, to understand how seasonal conditions shape yield, profitability, resource efficiency and risk. After cleaning missing values and treating outliers, the analysis applies statistical testing (one-way ANOVA), correlation analysis and a range of visualizations to compare performance across seasons, crops, irrigation methods and states.

The headline finding is a gap between production and profitability: while Kharif delivers the strongest yields and average profit, a majority of farms across all seasons are operating at a financial loss — and this loss rate worsens steadily from Kharif to Zaid. The report below sets out the methodology, key results and evidence-based recommendations that follow from the data.

1. Introduction & Problem Statement

Agricultural performance is shaped by seasonal variation in environmental conditions, farming practices, resource availability and market conditions. Raw agricultural data on its own does not reveal how performance changes across seasons or what patterns explain those differences. This project investigates seasonal differences in agricultural performance by identifying meaningful patterns, trends, relationships and variations within the available data — rather than performing a generic, unfocused analysis of the full dataset.

1.1 Objective

- Explore and clean 4,000 farm records across three seasons, eight states and eight crops.
- Examine how yield, cost, revenue, profit, water efficiency and disease/pest risk vary by season.
- Test whether observed seasonal differences are statistically significant.
- Compare performance across crops, irrigation methods and states within each season.
- Translate findings into evidence-based recommendations for seasonal agricultural planning.

1.2 Key Analytical Questions

- How does yield vary across seasons, and is the difference statistically significant?
- How do environmental conditions (rainfall, temperature) differ by season?
- How do cost, revenue and profit vary by season — and are farms actually profitable?
- Do specific crops perform consistently well or poorly across all seasons?
- Does irrigation method affect water-use efficiency, and does this vary by season?
- Is disease/pest risk related to seasonal environmental conditions?
- Which variables correlate most strongly with yield and profit?
- Are seasonal patterns consistent across different states?

2. Dataset & Methodology

The dataset contains 4,000 farm-level records with fields covering farm identity, seasonal and environmental conditions (rainfall, temperature, humidity, soil moisture), farming inputs (fertilizer, irrigation method, seed quality), and outcomes (yield, production, cost, revenue, profit, water usage and disease/pest risk).

2.1 Data Cleaning

- Missing values in Rainfall, Soil Moisture and Yield were imputed using the median for each Season + Crop combination, preserving genuine seasonal and crop-level patterns rather than flattening them with a global average.
- Yield and Water-Use Efficiency contained extreme outliers (max values far beyond realistic field levels) that were treated using IQR-based capping, retaining all 4,000 records while reducing distortion.

2.2 Tools & Techniques

- Python 3, Pandas & NumPy — data cleaning, transformation and aggregation
- Matplotlib & Seaborn — box plots, heatmaps, scatter plots and bar charts
- SciPy — one-way ANOVA hypothesis testing
- Jupyter Notebook — end-to-end documentation of the analysis

3. Key Findings

3.1 Seasonal Data Overview

The dataset skews toward Kharif, India's dominant monsoon-driven cropping season, followed by Rabi and the much smaller Zaid season.

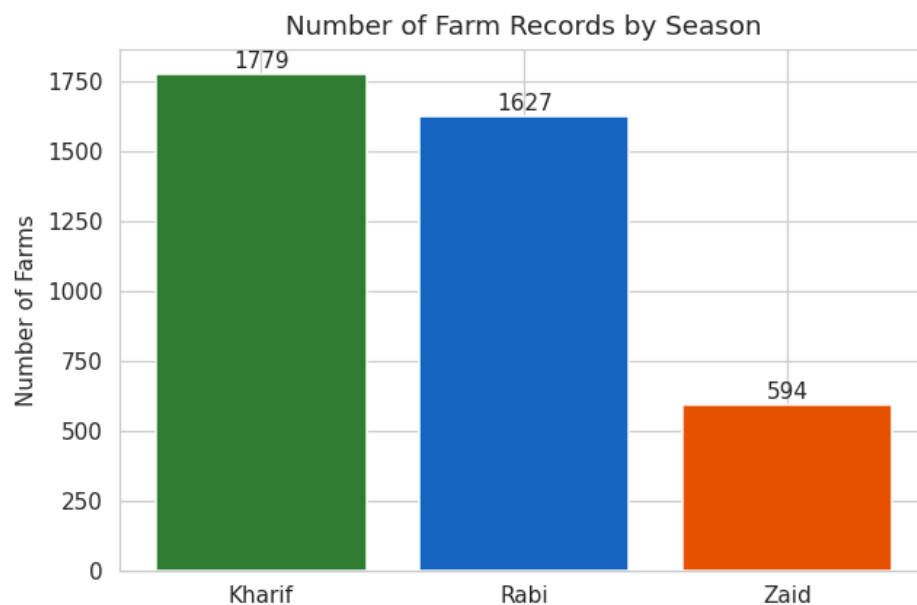


Figure 1. Number of farm records by season (Kharif 1,779 | Rabi 1,627 | Zaid 594).

3.2 Yield Across Seasons

Kharif records the highest average yield (2.26 t/ha), followed by Rabi (2.04 t/ha) and Zaid (1.81 t/ha). A one-way ANOVA test confirms this difference is statistically significant ($F = 25.05$, $p < 0.001$), meaning the seasonal effect on yield is real and not due to chance.

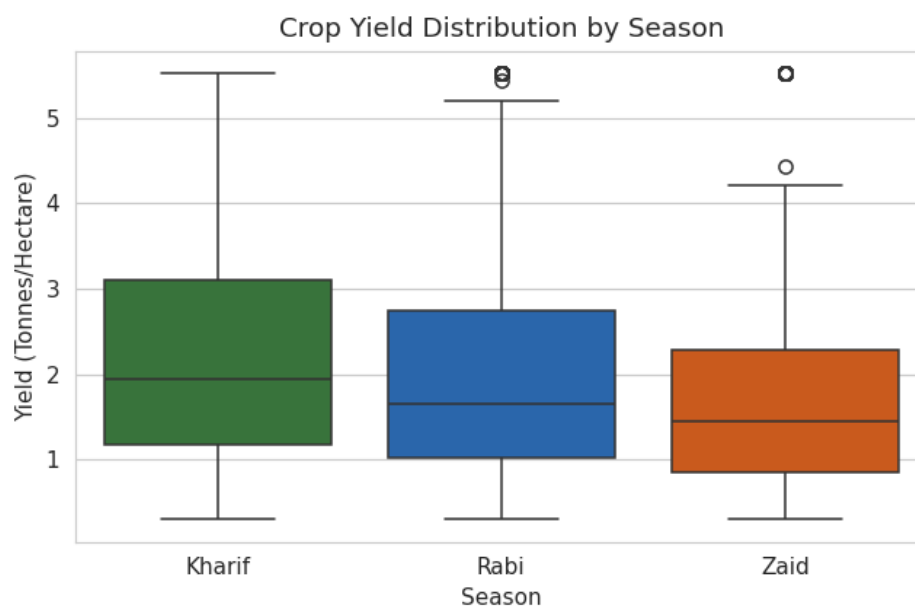


Figure 2. Yield distribution by season.

3.3 Environmental Conditions by Season

Kharif is the wettest season (852mm average rainfall); Zaid is the hottest and driest (31.0°C average temperature, 299mm rainfall). Rabi sits in between as the cooler winter season.

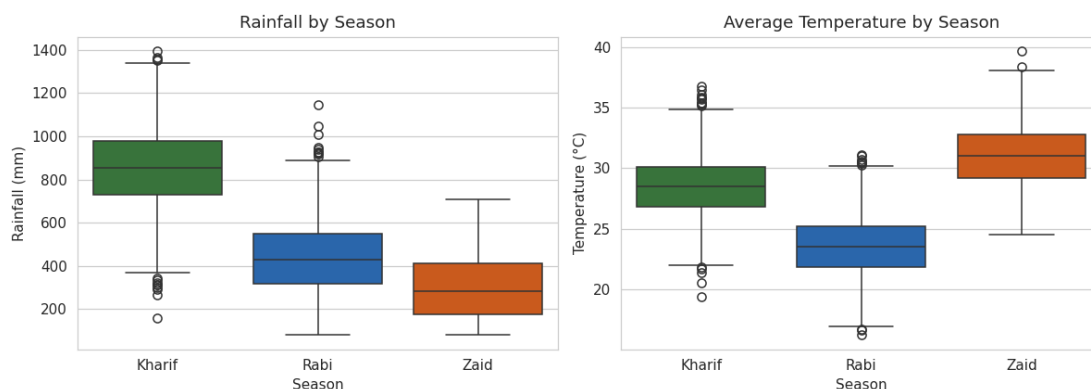


Figure 3. Rainfall and average temperature by season.

3.4 Economic Performance

Average profit falls sharply across the season cycle — from roughly ₹1.79 lakh per farm in Kharif to an average loss of ₹24,800 in Zaid — even though average cost stays broadly similar across seasons. A one-way ANOVA test confirms the difference in profit across seasons is also statistically significant ($p < 0.001$).

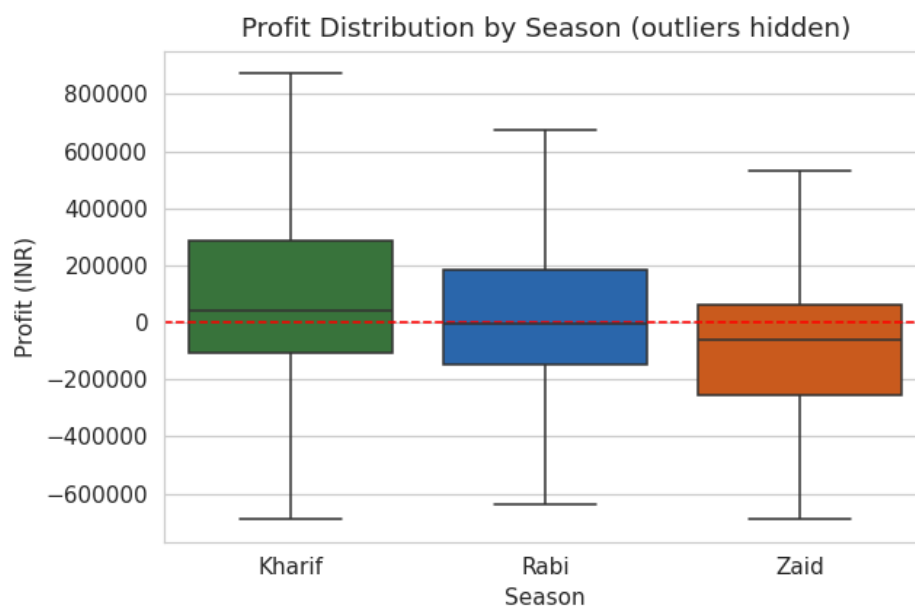


Figure 4. Profit distribution by season (outliers hidden for clarity).

3.5 Farm Profitability Risk

Perhaps the most important finding of this study: the share of loss-making farms rises steadily through the season cycle — from 42% in Kharif, to 51% in Rabi, to 65% in Zaid. This pattern is invisible if only average profit is examined.

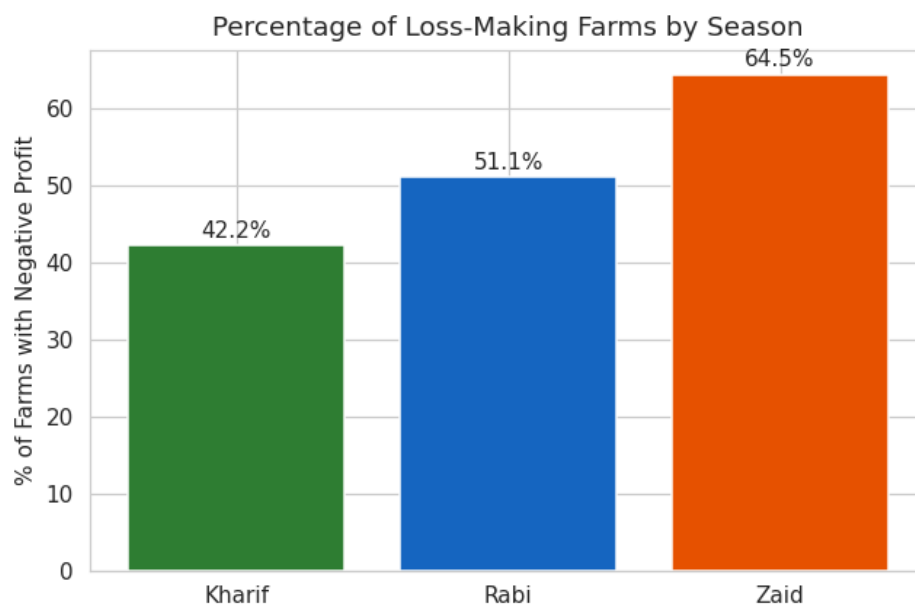


Figure 5. Percentage of loss-making farms by season.

3.6 Crop Performance & Risk Across Seasons

Crop choice matters as much as season. Sugarcane yield remains stable at roughly 5.45 t/ha in every season and has the lowest loss rate (8–15% of farms). Wheat, by contrast, is the highest-risk crop in every season, with 68–92% of farms loss-making despite reasonable yields — pointing to a cost or market-price problem rather than a production problem.

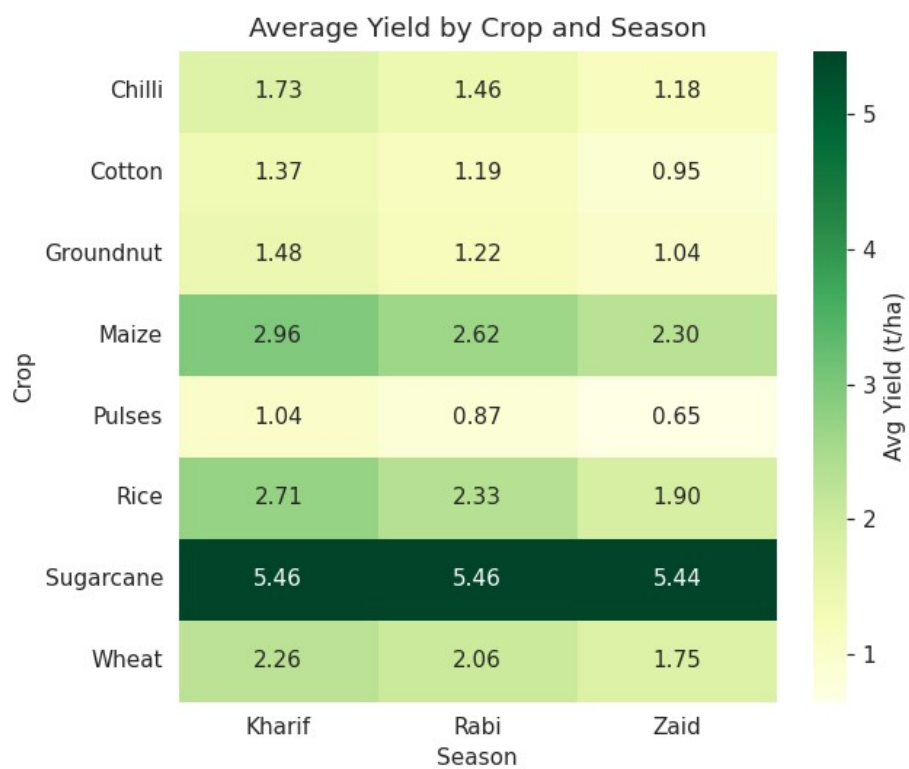


Figure 6. Average yield by crop and season.

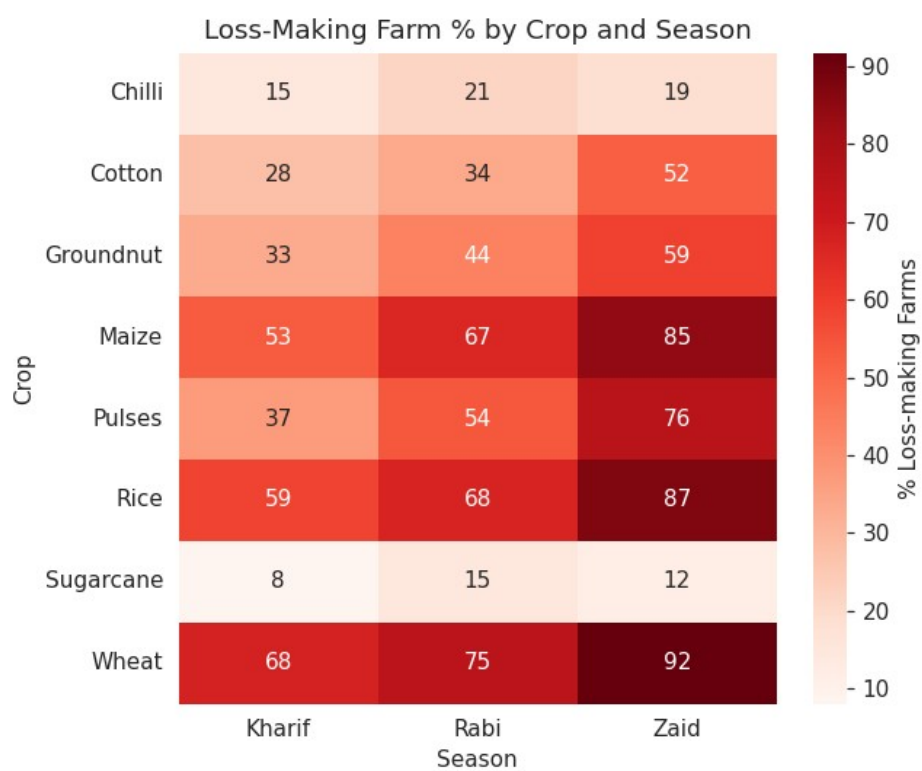


Figure 7. Loss-making farm percentage by crop and season.

3.7 Water-Use Efficiency by Irrigation Method

Irrigation method has a bigger effect on water efficiency than season does. Rainfed farming is the most water-efficient (5.19 tonnes produced per 1,000 m³ of water), while Flood irrigation — the most widely used method at roughly a third of farms — is the least efficient (2.67).

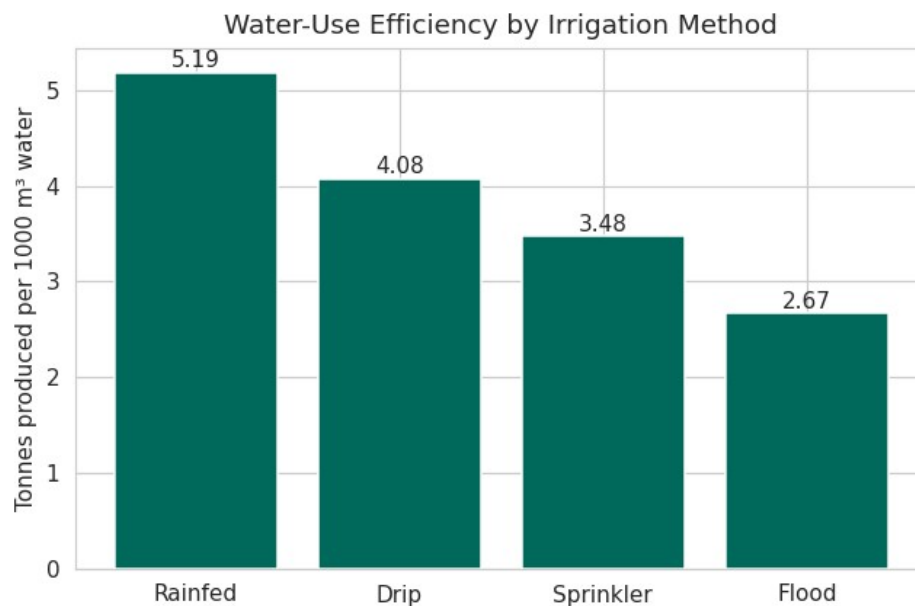


Figure 8. Water-use efficiency by irrigation method.

3.8 Disease / Pest Risk Patterns

Disease and pest risk is highest in Kharif (54.5%) and correlates strongly with rainfall ($r = 0.62$) and humidity ($r = 0.55$), confirming that wetter seasonal conditions drive higher disease pressure.

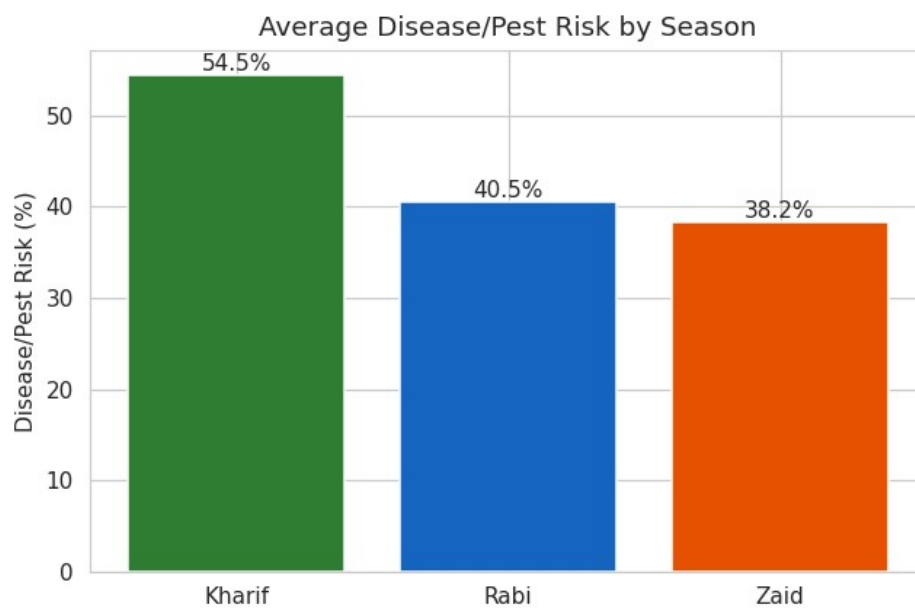


Figure 9. Average disease/pest risk by season.

3.9 Correlation Analysis

Yield and profit correlate only weakly with any single factor such as fertilizer use or rainfall ($r \approx 0.1$), suggesting outcomes are driven by a combination of crop type, irrigation method and market conditions rather than any one input.

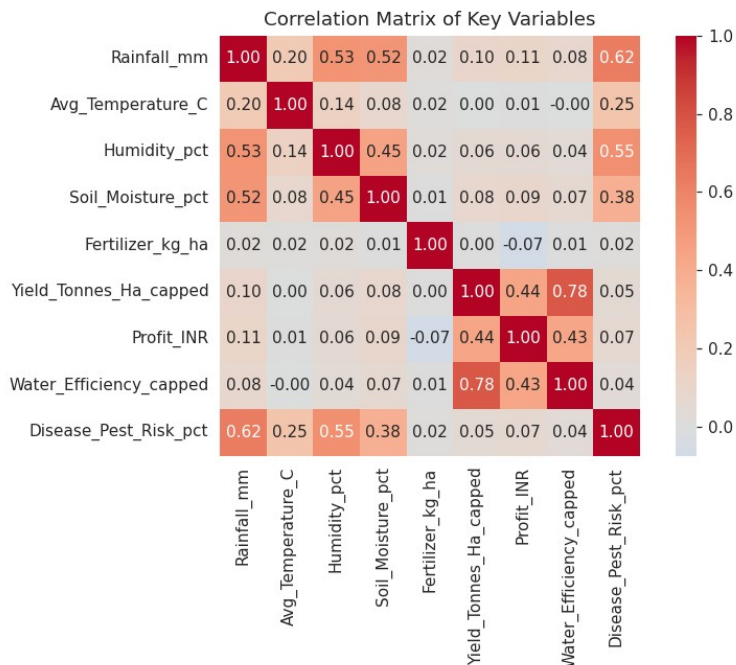


Figure 10. Correlation matrix of key numerical variables.

3.10 State-wise Seasonal Performance

Profitability also varies substantially by state, even after accounting for season. Punjab and Maharashtra show the highest average profit, while Andhra Pradesh and Madhya Pradesh trail behind — indicating that regional factors compound seasonal effects.

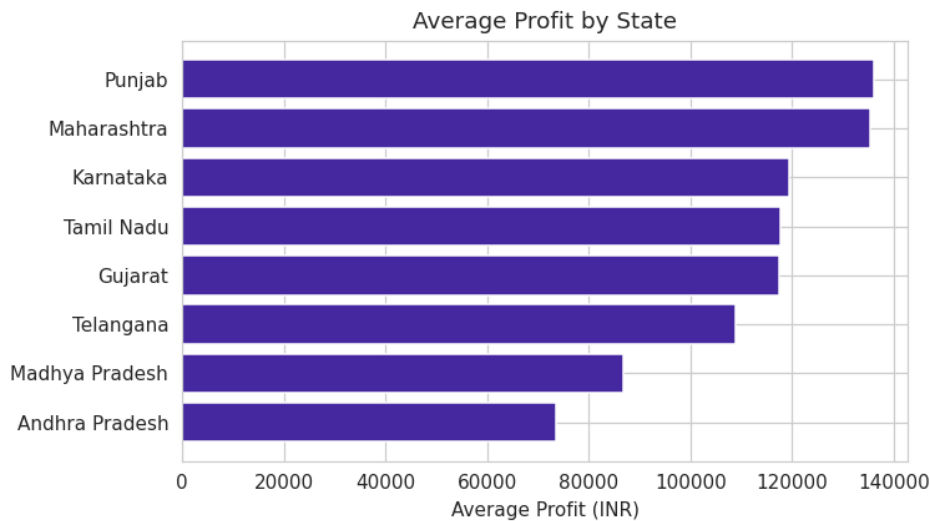


Figure 11. Average profit by state.

4. Key Insights

- Kharif is the strongest season overall: highest yield, highest profit, but also the highest disease/pest risk.
- Zaid is the weakest season: lowest yield and the only season with a negative average profit.
- Most farms are loss-making, and the situation worsens through the season cycle (42% → 51% → 65%).
- Yield and profit differences across seasons are statistically significant (ANOVA $p < 0.001$ for both).
- Crop choice matters as much as season — Sugarcane is stable and low-risk; Wheat is high-risk in every season.
- Irrigation method drives water efficiency more than season does — Rainfed is most efficient, Flood is least.
- Disease/pest risk rises with rainfall and humidity, explaining Kharif's elevated risk exposure.
- State-level profitability gaps persist even within the same season, pointing to regional factors beyond climate.

5. Recommendations

- Protect Kharif-season returns with stronger disease/pest management, given its highest risk exposure.
- Re-evaluate Zaid-season cultivation choices given its negative average profit and highest loss rate.
- Investigate Wheat's cost structure separately — its losses stem from cost/price, not yield.
- Promote diversification toward stable, low-risk crops like Sugarcane where conditions allow.
- Shift irrigation investment from Flood toward Drip or well-managed Rainfed systems.
- Target disease/pest advisories to high-humidity, high-rainfall periods, especially Kharif.
- Use state-specific benchmarks alongside seasonal ones, since profitability gaps persist within the same season.

6. Conclusion

This analysis confirms that season is a statistically significant driver of both yield and profitability, but it is not the only one — crop choice, irrigation method and location interact with seasonal conditions to shape outcomes. The most urgent finding is that a majority of farms are operating at a loss, with the situation worsening from Kharif to Zaid. Seasonal agricultural planning should therefore focus not just on maximizing yield, but on closing the gap between production outcomes and economic viability — particularly for Zaid-season and Wheat cultivation, where the data shows the greatest financial risk.

7. Future Scope

- Extend the analysis to real-time weather and IoT soil-sensor data for live seasonal monitoring.
- Build a machine learning model to forecast yield and profit ahead of each season.
- Develop an interactive dashboard for farmers and planners to explore seasonal trends.
- Add a crop-and-season recommendation engine based on a farm's location and soil profile.
- Incorporate market-price forecasting to strengthen the economic side of the analysis.

Project resources

Full Jupyter Notebook, dataset and presentation deck: <https://github.com/vaibhav2005-ssjoc/vois--main-project->